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Walking on air or hopping mad? Understanding the impact of emotions, sentiments and reactions on ratings in online customer reviews of mobile apps

Suparna Dhar [a], Indranil Bose [b,*]

[a] NSHM Knowledge Campus, 124, Basanta Lal Saha Rd, Tara Park, Behala, Kolkata 700053, West Bengal, India
[b] NEOMA Business School, 59 Rue Pierre Taittinger, Reims, 51100 Reims, France

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ABSTRACT

Online customer reviews (OCR) are an established source of competitive intelligence. Since OCR reflect perceptions of past users about product features and performance, prospective customers of mobile apps often resort to OCR before they decide to download an app. In addition to the textual content of the OCR, potential users of the app often consult the star ratings provided by the reviewers. In this paper, we determine if the star rating of mobile payment apps suitably captures the appropriate customer perception reflected in the textual content of the OCR for the payment apps. We conduct a three-dimensional analysis of 146,914 reviews from two popular mobile payment apps and uncover the sentiment and emotions captured in the review text, and the reactions in the emojis. Our analysis shows that the consolidated sentiment and the happiness emotion obtained from the OCR of mobile payment apps impacts the star ratings more strongly, than other emotions embedded in the text, and reactions in the emojis. The findings also hold good during the Covid-19 pandemic. We propose a novel perception score that captures the nuances in the OCR more effectively and suggest how mobile app providers can use it to arrange the OCR for better compatibility with content. The results of this research will be useful for mobile app providers that depend on OCR for popularizing their apps and will provide them a novel metric based on content, that can leverage OCR better to influence future downloads.

"When dealing with people, remember you are not dealing with creatures of logic, but creatures of emotion."

Dale Carnegie

1. Introduction

The use of the mobile internet and mobile applications (popularly referred to as the 'mobile app' or simply 'app') have seen a steep growth in recent years and is likely to continue in the coming years. In the competitive market of mobile apps, user experiences have played a critical role in determining market penetration of the app. App users have shared their experiences and opinions about the apps in online customer reviews (OCR). The OCR content and the star ratings of the apps have played a pivotal role in determining a prospective user's decision to download and use an app [1,2]. The market and the marketers have gained information on consumer reactions to an app by studying the rich content of OCR (content having text and emojis). Thousands of customer reviews have crowded the review pages of mobile apps. In the review pages, the reviews are generally sorted in reverse chronological order, or relevance, or star rating, or at random. These presentation arrangements have not considered the customer's perceptions in terms of emotions and sentiments expressed through the review. Fig. 1 depicts a sample screenshot with OCR data for apps. The figure depicts four reviews, one with a star rating of 5, two with star ratings of 4, and one with a star rating of 2. The two reviews "*Very nice app, useful for everything, sometimes getting small issues in the time of balance enquiry. Retry retry*" and "*Please add some offers of cash back every time it shows offers on rummy ace one etc which are useless*" share the same rating, but the former appears more positive than the latter. The star ratings do not distinguish between the positivity levels in the review content. The review "*Two years before I was completely …*" had a star rating of 2, but shows an

* Corresponding author.
E-mail address: indranil_bose@yahoo.com (I. Bose).

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element of positivity almost at par with the reviews having star rating of 4. This shows that the star rating has an element of subjectivity other than the customer's preference expressed through them. In the absence of a tool to measure the (positive) perceptions in the OCR content, the sites have arranged the reviews either chronologically or randomly. A mix of good, tepid, and bad (perceptions expressed in) reviews can confuse, frustrate, and even turn away a prospective consumer. A tool to methodically analyze feelings expressed through the OCR data will help marketers analyze and synthesize the data, and provide an option to organize the OCR data according to the preferences expressed by customers in the textual content.

Iyer & Kuksov [3] have found that the customer's affect signals product quality and it influences the decision to purchase a physical good. Vila & Kuster [4] have shown that at times, the customer's preference factor has outperformed the superior design factor in influencing the adoption of digital goods and services. This phenomenon is expected to extend to the adoption and use of mobile apps. Customer preferences are expressed through sentiments and emotions that influence human behavior and decision making [5]. Though the terms 'sentiments' and 'emotions' are often used synonymously, they are essentially different. Psychologists explain emotions as "powerful inner forces that affect our behavior and thoughts" [6]. The emotions are demonstrated through facial, verbal and written expressions. On the other hand, a sentiment is an opinion or a view. The Webster English dictionary defines sentiment as "an attitude, thought, or judgment prompted by feeling". Emotions are physical manifestations, and assessing the wide range of emotions involve a complex analysis, while sentiment is a mental condition that is bipolar.

The OCR contains a rich content (text and emojis) and star ratings. The unstructured nature of the rich content calls for the use of advanced data mining techniques. The star ratings provide objective information for easy analysis [7]. It may be argued that the star ratings summarize the customers' opinion as expressed in the textual content of the review [8,9]. However, the discrete 5-point scale of a star rating is inadequate to capture the wide range of opinion that is expressed in the content of the OCR, as evident from Fig. 1. This leads us to our first research question:

Is the star rating an effective measure of the customers' opinion expressed in the textual content of the OCR of mobile apps?

The usage of mobile apps in India has been an enigma. As per a report by Newzoo, India had 439 million smartphone users in 2020; one of the highest in the world [10]. Yet, the penetration of smartphones (per million population) in India is low compared to many other countries. Among the numerous apps used in India, the payments apps have gained a huge momentum in the last two years. Dilip Asbe, CEO and MD of National Payments Corporation of India has remarked, "With the increased adoption of mobile phones and greater penetration of the Internet, the awareness of digital payments started percolating into the grassroots levels of society" [11]. A higher adoption of payment apps can accelerate India's pursuit for a digital economy and financial inclusion. We conducted interviews with some mobile phone users in India to capture their opinion about using mobile apps for payments. Amit Singh, an educated young urban male user said, "I chose Google Pay. The reviews said it is convenient and secure." Anita Mondal, a rural homemaker said, "I know that people make payments from their mobile phones…but I never used it. I will ask my friends to suggest an app. I have read the reviews of PhonePe. They are so confusing.". Many people in emerging economies use the smartphone, but do not know how to use OCR successfully to find the right app that meets their needs. There is a need to devise methods to enable them to realize and utilize the full potential of the smartphone [9]. Reading through many reviews, and deciphering the content to make the decision to download an app is a daunting task for people like Anita Mondal. A visual tool that can summarize the perceptions expressed through the OCR can help such people leverage the review information for deciding whether to use an app.

There is a need for a tool that can measure the customer's opinion expressed through the OCR. The tool will allow marketers to gauge, monitor, and use this information for the purpose of marketing. For example, it will allow the marketer to compare the OCR, sort them according to their positivity/negativity, and conditionally select OCR satisfying predefined criteria of positivity/negativity. A summary analysis of the score will be useful to gauge the customers' opinion of the app and for comparative analysis of competitive apps. We endeavor to identify a continuous quantitative score that can ascertain the positivity of opinions expressed in the textual content of the OCR. We refer to this metric as 'perception score'. This leads us to our second research question:

How can we represent the opinion expressed in the textual content of the OCR of the mobile apps in the form of a continuous perception score?

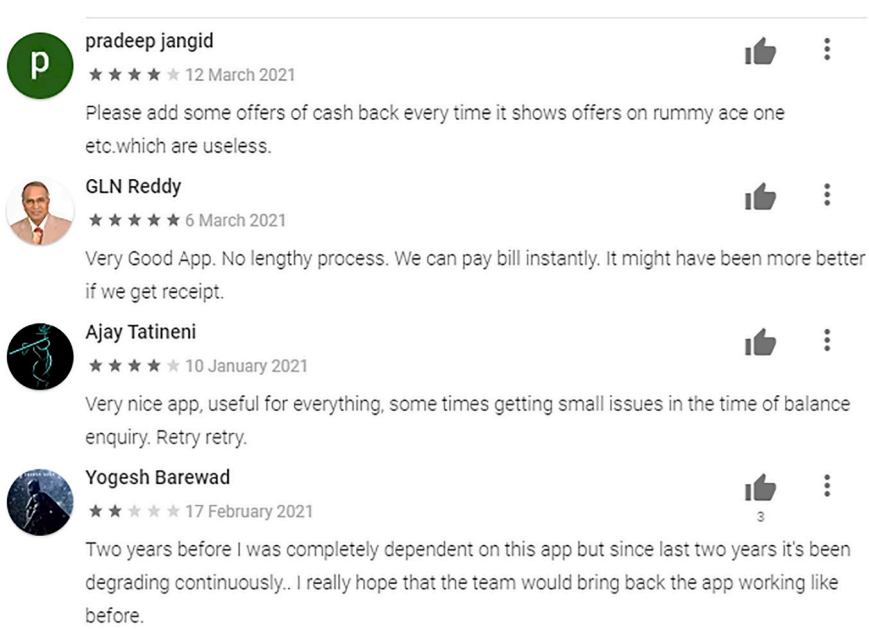

Fig. 1. A sample screenshot of the OCR data for a mobile payment app.

In this paper, we analyzed 146.914 reviews collected from the Google play store for two of the leading mobile payment apps (Google Pay and Phonepe) that are used in India. We used data mining techniques to identify the opinions expressed in the OCR of these mobile payment apps. We extracted sentiment polarity, scores for the eight basic emotions [6] from the textual content as well as positive and negative reactions from the emoji content. Our multivariate linear regression analysis showed that the opinions embedded in the content of the OCR are significantly associated with the star rating, though the explanatory power of the model is only 33.5%. We found that the consolidated sentiment and the happiness emotion obtained from the OCR of mobile payment apps impacts the star ratings more strongly, than other emotions embedded in the text, and reactions in the emojis. In order to represent the sentiments, emotions, and reactions in the textual content and emojis of the OCR more effectively through a continues score, we defined a perception score based on the regression analysis. Statistical validations showed that the score was a good representation of the customers' opinions in the OCR content for apps. We showed how the ordering of the reviews by the perception score can provide simpler, richer and summarized information from the textual content of the OCR. This has the potential to reduce the time and effort of customers who are scavenging for meaningful information from the OCR, to decide whether to download an app to serve their need.

We present the background information for this research in Section 2, the data collection and processing in Section 3, analysis and results in Section 4. We discuss our findings in Section 5 and provide our concluding remarks in Section 6.

2. Background

2.1. Determinants of app download and usage continuance

The adoption of mobile apps has evoked academic interest in the past. Scholars have used the technology acceptance model, theory of reasoned action, and diffusion of innovation, to examine the antecedents for the adoption of mobile apps [12,13]. Some scholars have focused on app specific strategies, such as privacy concerns, and licensing and subscription in determining the intention to adopt an app and continue to use it [2,14]. Social influences and subjective norms have significantly impacted the customer's intention to use mobile apps [15]. In this regard, the customer reviews of apps have acted as a potential source of expert information and have influenced the decision of the customers to install and use the apps [1,14]. The review data have indicated the customers' value, satisfaction, and loyalty for the app and their recommendations for prospective users [16].

2.2. Online customer reviews (OCR)

Online customer reviews serve as a critical source of feedback for products and services [17]. The customer reviews reflect the customers' brand-related opinion and influence the perception of the uninformed online consumers [5,7]. Marketers have used the OCR data to evaluate the role of brand perceptions and customer satisfaction [18], in driving online sales of products [19]. Several apps that include a variety of functionalities have crowded the app store and the play store. In the competitive market of mobile apps, new users have depended on the star ratings of the apps to make a decision to download it [1,2]. The content of the OCR and the star rating have provided cues about characteristics of apps, evaluations by users, and customers' preferences [15]. This makes OCR a goldmine for marketers for harnessing competitive intelligence [17,20].

Past studies have found that positive reviews positively influenced consumers' product choices [21]. Some scholars have extracted latent topics and textual parameters from the textual content of the OCR to examine the customer satisfaction and the product recommendation behavior [22,23]. Researchers have studied the influence of sentiments and emotions in the textual content of the OCR on purchase decisions [5,24]. Extant research has reported that visual cues in the OCR have played a critical role in determining its impact [25].

The star ratings by the customer have provided objective cues about the users' feedback. Extant studies have found an association between the star ratings in OCR content and the customers' intention to continue the usage of mobile apps [26,27]. The OCR of mobile apps have also provided cues to developers about improvements of the design and performance [28,29]. Star ratings of an app serve as a form of electronic word-of-mouth that influences the decision to install the app [30]. Since the star ratings are not a numeric representation of the sentiments expressed in OCR text [31], the association between the emotional content of the OCR and the star rating have varied for different product categories [7].

2.3. Opinions expressed in the OCR

In the offline scenario, retailers and restauranteurs focus on creating an ambience to enhance the customers' opinion [3]. In the online scenario, marketers have endeavored to improve the helpfulness of websites by including OCR that contain customers' opinion [4]. The opinion in the OCR is expected to influence other customers. Human opinions are complex, personal, internal, and often confusing [6]. Developments in the field of text and data mining have used techniques to extract sentiment polarity (positive, negative), and emotions from unstructured expressions [5]. Information systems scholars have used the sentiments and emotions expressed in the textual content of the OCR to study their impact [24,32]. While most scholars have studied either the sentiment in OCR [31] or emotions [32], some have studied the sentiments and emotions expressed in OCR in unison. Chatterjee [5] has extracted the sentiments, emotions, and other qualitative information from OCR to analyze their influence on star ratings for airlines.

The appraisal theory has suggested that the cognitive evaluation of the situation may result in emotional reaction [33]. In psychological studies that describe various human emotions, there is a lack of consistency in the classification of emotions [33]. The psycho-evolutionary theory has identified anger, anticipation, disgust, fear, joy, sadness, surprise, and trust as the eight basic emotions [6]. These basic emotions include four positive emotions (anticipation, joy, surprise, and trust) and four negative emotions (anger, fear, disgust, and sadness). The eight basic emotions expressed in reviews of products have affected the performance of the product in a variety of ways [17]. For example, the joy expressed in the OCR has indicated customers' delight with the product and service quality; which in turn has led to their acceptance and continued use of the product [34,35]. On the other hand, discontent expressed in the OCR has often indicated an adverse experience and has led to lower adoption or discontinuance of an app. The study of emotions in OCR has been of interest to scholars of information systems. Researchers have used a variety of techniques to extract emotions and sentiments from reviews [36,37]. They have analyzed the relationship between emotions in the OCR and their influence on perceived helpfulness, and other business consequences [8,17,32].

Table 1 provides a summary of the extant literature.

From Table 1 we can observe that research on the usage of apps have often used data extracted from their OCR. However, several gaps exist in this line of research. Extant literature have reported a difference in the association between the emotional content of the OCR and their corresponding star ratings for different industrial sectors [7] such as airlines and tourism [5,41]. To the best of our knowledge, there is no such study in the case of mobile apps. At the same time, although scholars have used text mining techniques to extract sentiments and emotions from customer reviews [24,36], there are only a few studies that have focused on extracting sentiments and emotions expressed on OCR for apps. Emojis or emoticons have become an integral part of communication on digital media [42] and there is a need to include the customers' reactions expressed through the emojis in the OCR. Again, there are no

Table 1
Summary of extant literature.

Paper	Description	Data source
A. Online customer reviews		
[38]	Studied the impact of various aspects of stories embedded in the OCR about the service-related expectations of customers.	OCR data from Yelp
[39]	Studied customers' intention to continue using a platform based on the OCR.	Survey of users of TripAdvisor
[40]	Used machine learning techniques to predict ratings from attributes and topics in the OCR.	OCR data from Airbnb
[22]	Applied text mining to extract topics from the OCR and examined their impact on customers' satisfaction with products and services in hospitality.	OCR data from booking.com
[23]	Mined the OCR for hotels to identify latent topics and identified their influence on product recommendations.	OCR data from TripAdvisor
[41]	Studied the association between customer satisfaction and customers' involvement in the review community.	OCR data from TripAdvisor
B. Online customer reviews for mobile apps		
[27]	Studied the interactions between app store strategy, and the hedonic and utilitarian value of mobile apps in influencing the star rating of the app.	Data from Play Store
[28]	Studied the effect of the OCR of mobile apps on the design of the app.	Data from Applause.com
[29]	Applied big data analytics to investigate the impact of the OCR on customer agility and product performance.	Data from App Store
C. Sentiments and emotions in online customer reviews		
[5]	Examined the influence of sentiments, emotions, and topics from the OCR on star ratings for airlines.	Data from airlinequality.com
[17]	Examined the association between emotions in the OCR and the helpfulness of the review.	OCR data from Amazon
[24]	Studied the influence of emotional content (pleasantness) of the OCR on customers' purchase decisions.	Experiment
[32]	Examined the relationship between negative emotions such as anger, fear, and sadness and the perceived helpfulness of the OCR.	OCR data from Amazon
[7]	Applied text mining techniques to extract emotions in the OCR and study their influence on star ratings.	OCR data from Amazon
[8]	Studied the perceived helpfulness of emotions embedded in the OCR.	Experiment and from Yahoo

studies that have assessed the impact of reactions expressed through emojis on the star ratings of the OCR. Finally, there exists a gap in the identification of a tool that can score the OCR based on the sentiments and emotional content in them. Our research in this paper strives to fulfill these gaps in the extant research.

3. Data collection and processing

The use of mobile payment apps has allowed a reduction in the use of cash, offered convenience and speed, and allowed secure transfer of money to individual payees and traders such as restaurants and retailers. India is one of the leading markets for mobile payments. The United Payment Interface (UPI) is a payment system in India that has connected banks and financial institutions to facilitate person-to-person and person-to-merchant money transfer using apps. The UPI was launched in April 2016. Within the first 5 years of its launch, the total value of transactions on the UPI have accounted for 15% of India's GDP (for the fiscal year 2019–20). UPI transactions have constituted approximately 59% of all digital payment transactions in India for the financial year 2020–21 until January 2021. Business leaders such as Mark Zuckerberg, the CEO of Facebook, and Sundar Pichai, the CEO of Google, have hailed the success of the UPI. Phonepe and Google Pay are two highly popular payment apps in the Indian market that account for more than 81% of the transaction volume during a month on the UPI. The Phonepe app has recorded 1199.5 million transactions, and Google Pay has recorded 957 million transactions in March 2021 [43]. The immense popularity of these two payment apps has prompted us to use the data obtained from the OCR of these two apps in our research. We collected 146,914 OCR between 1st January 2019 and 15th March 2021 for these two leading UPI-based payment apps from the Google play store using Python API. In the collected OCR data, 61.85% of the reviews came from Phonepe and the remaining 38.15% came from Google Pay. Fig. 2 shows the distribution of the OCR from 2019 to 2021.

3.1. Data analysis techniques

Our literature review showed that extant research on the topic extracted sentiments and emotions from the textual content of the OCR. Some papers conducted a single-dimensional analysis [17,37] and others a two-dimensional analysis [7,44] of the textual data. These papers focus on the textual content of the OCR but do not consider the emojis in the OCR. We performed a three-dimensional analysis by extracting the sentiment and emotions embedded in the OCR text, as well as reactions represented in the emojis. The sentiments analysis was used to extract the positive-negative sentiment polarity in the OCR and this constituted the first dimension ($\text{Sentiment}_{\text{Text}}$). The emotion analysis was performed to extract the emotion scores for the eight basic emotions expressed by the textual content of the OCR and this represented the second dimension ($\text{Emotion}_{\text{Text}}$). The emoji analysis was performed to extract the positive-negative reactions expressed by the emojis and this accounted for the third dimension ($\text{Reactions}_{\text{Emoji}}$). Fig. 3 depicts the approach that we adopted for data analysis.

We performed multivariate linear regression analysis for determining the association between sentiments, emotions, and reaction with the star rating of the OCR. Following standard practice, we used the sentiments and emotions from the text and reactions from emojis as independent variables [5] and the star rating of the app in the OCR as the dependent variable. Since prior research has shown that structural features such as the length of the OCR are significantly associated with the star rating [41], we used the length of the OCR (i.e., number of words) as a control variable. Since the number of words can vary widely (less than 10 to more than 100), we used a logarithmic (base 10) transformation to deflate the measurement. Additionally, we also considered the number of emojis as the second control variable. We have taken the app as the third control variable to control for app-specific differences if any. We have tested the independence of the observations using the Durbin-Watson statistic, homoscedasticity by inspection of the scatter plot of errors, multicollinearity by comparing the variance inflation factor, outliers by the inspection of the scatter plot of standardized residuals on standardized predicted, and the normality of residuals by inspection of the normal probability plot.

3.2. Sentiment analysis

Sentiment analysis involves the identification of the positive-negative polarity of the text [44]. We used the NLTK sentiment analyzer to extract the sentiment polarity from the textual data as a single number in the range of −1 to +1. The NLTK uses a supervised machine learning approach to analyze sentiments in textual data [37,41,45]. The results of the analysis were manually verified by the two authors and the interrater reliability was found to be high (kappa = 0.85) [46]. Any differences in the verification process were resolved through discussion.

The NLTK sentiment analyzer API returned 4 sentiment scores – positive, negative, neutral, and consolidated. The consolidated score ranged between −1 and + 1, where the negative score indicated negative sentiment and the positive score indicated positive sentiment. The magnitude represented the intensity of the positive or the negative

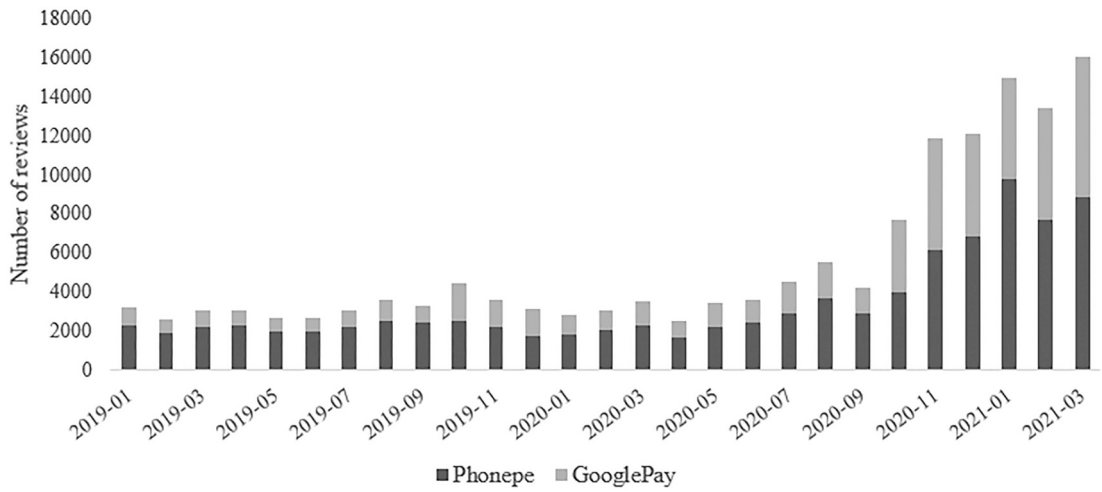

Fig. 2. Distribution of the OCR data over the years (2019–2021).

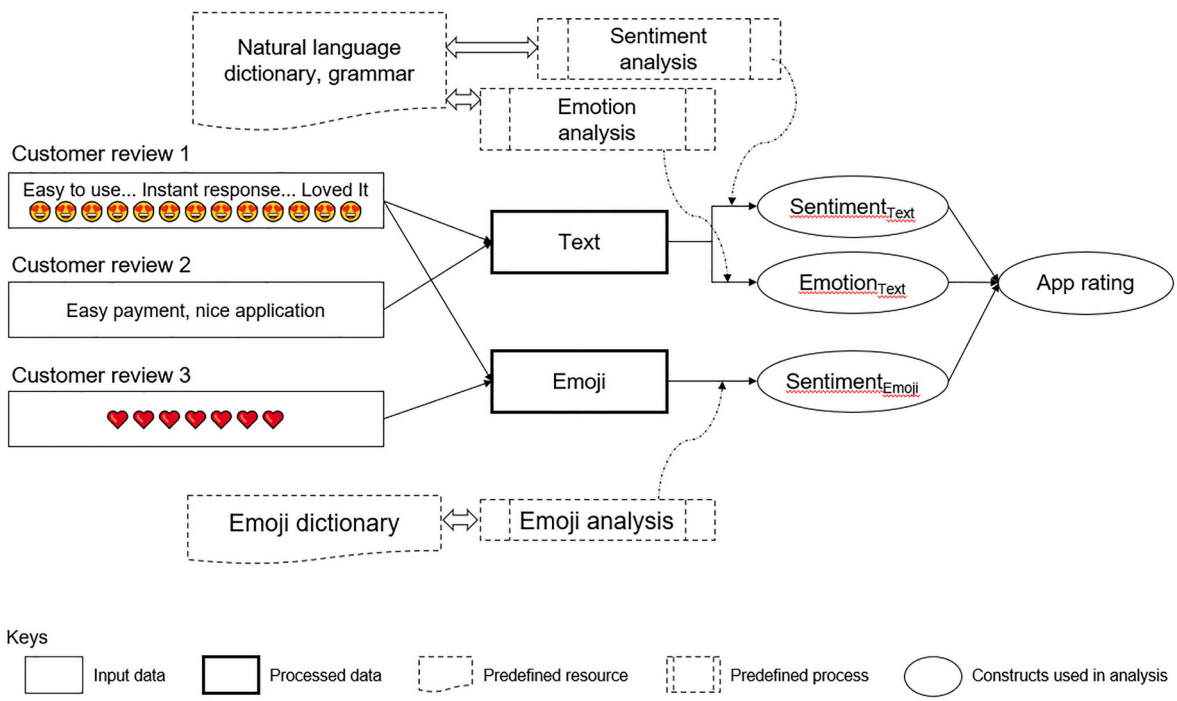

Fig. 3. A process diagram for analysis of the OCR of the mobile payment apps.

sentiment. We used the consolidated score in our analysis. The sentiment analysis scored the reviews "*Safe, secure, easy and amazing*" and "*nice and perfect*" as highly positive; "sad" and "*bad connection, server problems*" as highly negative. The reviews "*One of the best app to transfer the money or any recharge like mobile*" and "*Very nice app, is a good platform for all shop keeper and business to accept payment in digital method, so easy…*" received a score higher than 0.9; and the reviews "*The worst app and the worst team support that I have seen*" and "*The App is good with more options & features, but most of the time the UPI transaction fails*" scored less than −0.9 as the consolidated sentiment score.

3.3. Emotion analysis

Emotion analysis involved the identification of emotions in textual data. We used the NRC word-emotion association lexicon to identify the latent basic emotions in the OCR [17]. This lexicon has annotated 14,182 English words with eight basic emotion classes (anger, anticipation, disgust, fear, happiness, sadness, surprise, and trust) defined by the psycho-evolutionary theory. We created an emotion matrix $\{e_{i,j}\}$, where $e_{i,j}$ was encoded as 1 if emotion i was annotated with word j, and 0 otherwise. We vectorized the words in text as $\{w_i\}$ where w_i represented the number of occurrences of the ith word in the text. The emotion scores were derived by $\boldsymbol{w}^T\boldsymbol{e}$. The results of the analysis were manually validated by the two authors and the interrater reliability was found to be high (kappa = 0.78). The differences were resolved through discussion.

The emotion analysis scored each review for the 8 basic emotions. Table 2 provides a snapshot of reviews that scored high on various dimensions of emotion.

3.4. Emoji analysis

Emojis or emoticons have become a norm in digital communication [42]. The symbolic language offers a powerful mechanism for expressing

Table 2
Sample OCR with high emotion scores.

Emotion	Sample OCR
Anger	"*I need my amount. What will I make of your clarifications? If you don't have any authority to save our money then why are you mediating like that?...*"
Anticipation	"Till now I don't have any problem with google pay."
Disgust	"*Fully ridiculous, disgusting, worst services, always server & network issues…*"
Fear	"*Whenever we try to do a payment first the amount gets debited from the account and god knows when it will be credited to the recipient account…*"
Joy	"*This app is very cool n user-friendly. Safely money transactions can be made through this app…*"
Sadness	"*After the update, this app has lost all its credibility. Please stop using this useless app asap!...*"
Surprise	"*I lost my phone in an accident and was able to download the app on the new one and retrieve all the info. Great app and very smooth and It's good and I like how everything is working…*"
Trust	"*app is working excellent & process is very easy .... improved service is helpful to society...*"

thoughts, feelings, and preferences. We used the python 'emoji' library to identify the emojis and classify the emojis as having positive, negative, and neutral reactions. The emoji library provides positive, and negative scores (a positive integer proportional to the reaction polarity for the emoji) for each emoji. For example, in the emoji library the positive reaction score for the emoji "😂" was 6845 and its negative reaction score was 3614. We generated the reaction matrix for the emojis $\{r_{i,j}\}$, where $r_{i,j}$ where i represented the ith emoji, j was assigned 0 for positive score and 1 for the negative score, and $r_{i,j}$was assigned the reaction score. We vectorized the emojis in text as $\{v_i\}$ where v_i contained the number of occurrences of the ith emoji in the text. The reaction scores were derived from $\boldsymbol{v}^T\boldsymbol{r}$. The results of the analysis were manually validated independently by the two authors. The interrater reliability was found to be very high (kappa = 0.82). The differences were resolved through discussion.

The analysis resulted in the score for positive, negative, and neutral reactions in emojis for each review. The scores were positive integers. We transformed the scores to fall within a range of 0 and 3. On the transformed scale, score 0 represented low reaction and 3 represented high reaction. The review "*Easy payment app ..... I am happy* 😊😊😄😄" scored high on the positive reaction in emojis, and the review "*Cannot able to login with mobile number* 😭😭😭 *but after uninstalling and installing after 5 to 6 time it worked* 😂😂" scored high on negative reaction in emojis.

4. Analysis and results

The word cloud analysis performed and shown in Fig. 4(a), demonstrated a mix of positive ('best', 'good', 'easy use') and negative ('worst experience', 'disappoint', 'transaction fail') opinions in the OCR data. We annotated the 100 most frequently used keywords into positive, negative, and neutral. We performed the Mann Whitney *U* test to compare the distribution of positive and negative words in the list. The test yielded no significant difference in the distribution of positive and negative words. Fig. 4(b) presents the emojicloud drawn from the emojis in the OCR data. The emojicloud featured a higher proportion of negative reactions. This justified the need for the third-dimension of our analysis.

We cleaned the data to remove OCRs having non-English text. Researchers analyzing the star ratings for product reviews have reported a J-shaped curve representing the relationship between average number of reviews and star ratings [7]. The phenomenon is explained by a consumer bias for extreme ratings [47]. For the regression analysis, we used stratified random sampling without replacement from the resultant data to get a homogenous mix of star ratings. Table 3 presents the descriptive statistics and correlations for the independent variables. Fig. 5 presents the mapping of opinions across the discrete star ratings (1 to 5 stars). The positive reaction in emojis showed a J-shaped curve. The consolidated sentiment and the negative reaction in emojis showed a linear trend. The various components of emotions showed varied levels of linearity in their gradient. Most of the curves showed an elbow at the star rating of 3. A linear regression analysis is performed to test the relationship of the independent variables with star ratings. Table 4 presents the results of the multivariate regression analysis. Model 2 presents the linear regression model showing the association between sentiments, emotions, and reactions with the star rating. The overall model was significant, and the regression coefficients for the independent variables were significant. However, the model explained 32.2% of the variability.

Natural language is complex. There are wide variations in the use of words and tones. Extant literature has reported difficulties in identifying the tone of a message. For example, the identification of sarcasm is difficult when processing natural language [48]. In our analysis, review comments such as "*It's too convenient app to pay or transfer money to any account ..?*" scored high on the positive emotions and low on the negative emotions, and received a negative consolidated sentiment score. Sarcasm typically involves positive words with a negative sentiment. To address the problem with the tone in natural language processing, we used the consolidated sentiment as a moderator (model 3). This improved the performance of our model by 1.3%.

Fig. 4. (a) Wordcloud formed for the textual data of the OCR (b) Emojicloud formed for the emojis of the OCR.

Table 3
Descriptive statistics.

	Mean	Std. dev.	Range	Correlation coefficients											
				Positive reaction in emojis	Negative reaction in emojis	Anger	Anticipation	Disgust	Fear	Happiness	Sadness	Surprise	Trust	Consolidated sentiment
Positive reaction in emojis	42.76	589.787	Integer in [0,41,070]	1										
Negative reaction in emojis	27.71	379.035	Integer in [0,21,708]	0.84	1									
Anger	0.36	0.612	Integer in [0,6]	0.022	0.038	1								
Anticipation	0.99	1.030	Integer in [0,9]	0.033	0.034	0.422	1							
Disgust	0.15	0.407	Integer in [0,4]	0.012	0.031	0.53	0.129	1						
Fear	0.30	0.564	Integer in [0,6]	0.012	0.025	0.501	0.234	0.483	1					
Happiness	0.77	0.876	Integer in [0,8]	0.031	0.022	0.409	0.761	0.059	0.16	1				
Sadness	0.34	0.616	Integer in [0,6]	0.007	0.029	0.41	0.139	0.509	0.666	0.048	1			
Surprise	0.47	0.644	Integer in [0,5]	0.021	0.015	0.404	0.626	0.055	0.122	0.708	0.071	1		
Trust	1.28	1.273	Integer in [0,13]	0.028	0.03	0.422	0.633	0.132	0.231	0.715	0.18	0.526	1	
Consolidated sentiment	0.120	0.523	[−1,1]	0.017	−0.022	−0.242	0.034	−0.293	−0.283	0.203	−0.355	0.129	0.025	1

4.1. Perception score

Given the moderate explanatory power of model 3, we decided to determine a new metric called 'perception score' that would capture the emptions, sentiments, and reactions better than the star ratings. We used model 3 to generate this perception score. A histogram of the predicted perception scores from model 3 showed a bell-shaped distribution with scores between 0 and 5. Fig. 6(a) depicts the histogram for predicted scores. To adjust for any outliers, we calculated perception score so that the values would lie between 0 and 5.

$$Perception\ score = \begin{cases} predicted\ score\ if\ predicted\ score \in [0,1] \\ 0\ if\ predicted\ score < 0 \\ 5\ if\ predicted\ score > 5 \end{cases} \quad (1)$$

A mapping of the perception score to the star rating showed a near linear relationship, with slight flattening at the extremities. Fig. 6(b) depicts the mapping.

To validate the perception score, we compared the mean squared distance (MSD) between the perception score and the constituent perceptions with that between the star ratings and the constituent perceptions. We repeated this computation with the mean absolute deviation (MAD). Table 5 presents the comparison of the mean squared distances and the mean absolute deviations. The analysis showed that the MSD and MAD for the perception score was consistently lower than that for the star ratings. For example, the $MSD_{Perception\ score}$ for anger was 7.79 and it was lower than the MSD_{Rating} for anger which was 8.879. It demonstrated a higher efficacy of the perception score in presenting the level of anger in a review than the star rating.

The perception scores were manually verified independently by the authors. The interrater reliability was found to be very high (kappa = 0.83). Table 6 presents a sample of the OCR data showing their star ratings and perception scores.

4.2. Robustness check: The impact of COVID-19

The timeline of our analysis covered the COVID-19 pandemic. There were reports of severe emotional distress during this period. An ANOVA showed significant differences in the opinions expressed in the OCR during the pandemic (January 2020 to March 2021) compared to the earlier period (January 2019 to December 2019). To check the robustness of our model, we repeated the regression analysis with data limited to the COVID-19 period (model 4). We also repeated the analysis with data for a single app (model 5). Table 7 presents the results of this analysis. The direction and the significance of the main effects remained unchanged. The direction for the significance of the interaction terms remained unchanged, with minor exceptions. This supported the robustness of our proposed model.

5. Discussion

The first part of our analysis showed that the customers' opinions were significant associated with the star ratings, but the explanatory power of the model was low. The star ratings suffered from the limitation of being a 5-point discrete scale and also did not reflect the opinions in the OCR very strongly. The plotting of emotions for the 5-point star ratings showed different gradients for the eight basic emotions. The positive emotions showed more erratic curvature compared to the negative emotions. The consolidated sentiment from the OCR showed the highest impact ($\beta = 0.486$***) on star rating, followed by emotions in the text and reactions in the emojis. Among the emotions, happiness showed the strongest influence ($\beta = 0.182$***) on star rating. Three of the basic emotions (fear, happiness, and surprise) showed a positive influence on star ratings. The remaining five basic emotions (anger, anticipation, disgust, sadness, and trust) showed a negative influence on star ratings. The positive reaction in emojis showed a positive influence ($\beta = 0.057$***) and the negative reaction in emojis showed a negative

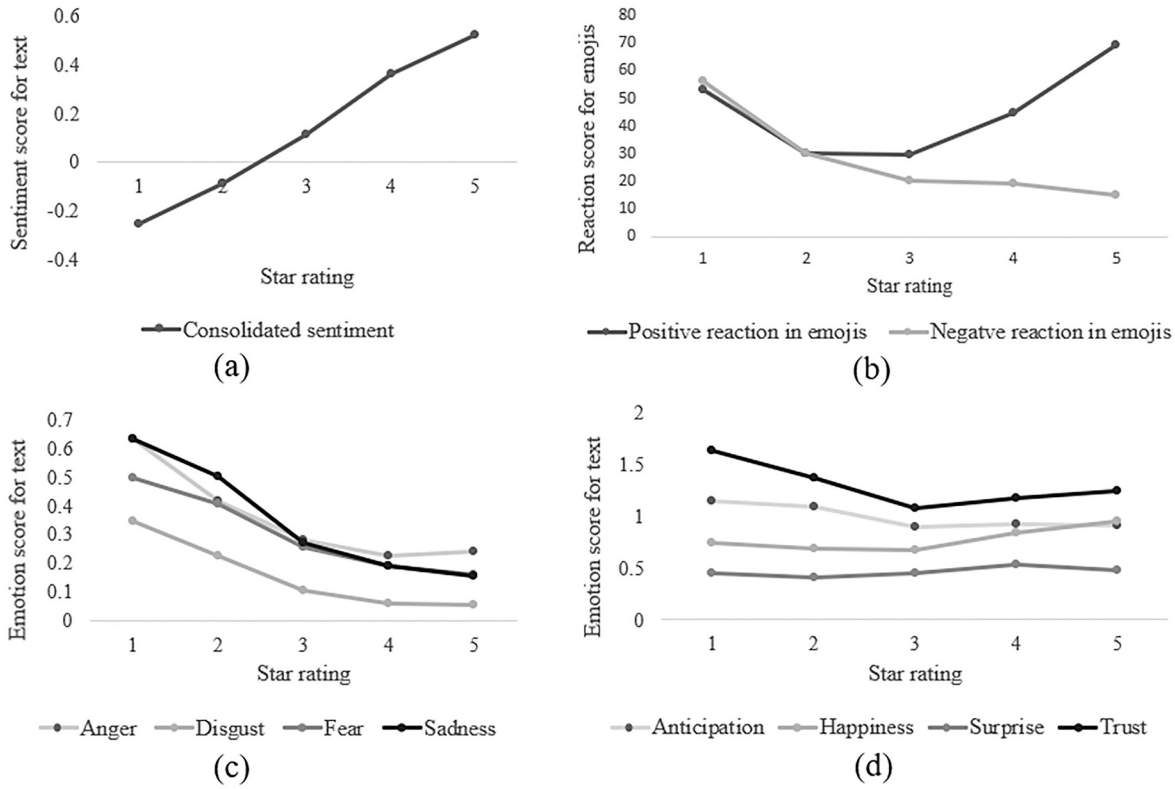

Fig. 5. The relationship of (a) $\text{Sentiment}_{\text{Text}}$, (b) $\text{Reactions}_{\text{Emoji}}$, (c) $\text{Emotion}_{\text{Text}}$ for positive emotion, and (d) $\text{Emotion}_{\text{Text}}$ for negative emotion, with star ratings.

influence ($\beta = -0.069$***) on star ratings. The consolidated sentiments moderated the associations between emotions in text and reactions in emojis on the star rating (model 3). Past studies that have analyzed and explained the star ratings extracted from the OCR have reported a gap between the content of the review and star ratings [5,41]. Our observations matched the findings of these studies.

In the second part of the analysis, we defined a perception score that reflected the sentiments and emotions in text and reactions in emojis more closely than the star ratings. Tables 5 and 6 demonstrated that the perception score is a better measure of the opinions expressed in the OCR. The perception score will enable marketers to gauge, monitor, and track consumer opinions about an app. Marketers will be able to score individual reviews and sort them and identify the top-ranking positive reviews for marketing campaigns on the basis of the perception score. Fig. 7 presents a proposed sorting of reviews (the original arrangement for these reviews is shown in Fig. 1) using the perception score. It is observed from Fig. 7 that the perception score has clearly demarcated the differences in the perceptions expressed in the two reviews. The review "*Very nice app, useful for everything…*" with a perception score of 3.718and star rating of 4 is ranked higher that the review "*Very Good App. No lengthy process…*" with a perception score of 3.390 and star rating of 5 for its strongly positive tone. Extant literature have shown that the sentiment polarity is a better predictor of sales compared to star ratings [31]. In line with that, the perception score, an embodiment of the sentiment and other components of the opinion, seems to be differentiating between the content of the reviews better than the star rating.

5.1. Theoretical contributions

Extant literature has expounded that the association between opinions in the OCR and the star ratings vary by industry [7]. Past studies have examined this association for tourism and airlines [5,41]. This paper made a unique contribution by analyzing the association between opinions expressed in the OCR and star rating for the mobile app industry. This paper contributed to the literature that have studied the social influence generated through the OCR. Researchers have text-mining to extract sentiment and emotions from large-scale OCR data for e-commerce [17,37]. Such an approach has ignored the information richness introduced by using emojis. This paper used a three-dimensional data mining approach to extract sentiments and emotions from the text and reactions from the emojis in the OCR for the mobile payment apps. It made a unique contribution by extracting reactions from emojis. The analysis stressed the media richness introduced by the emojis, and the consequent need to include emojis in research related to opinion mining. These findings supported the observation that social influences and subjective norms have a profound effect on the intention to use mobile apps [15].

Extant research has reported a J-shaped curve mapping the relationship between the number of reviews and associated star ratings [7,47]. In our exploratory analysis, we found that the positive reactions in emojis in the OCR followed a J-shaped curve when plotted against the star ratings. This indicated a higher use of positive reactions for extreme ratings. Possibly, the 5 star rating with a high level of positive reaction in emojis indicated positive opinion, and the 1 star rating with high level of positive reaction in emojis indicated sarcasm [48]. This shows that it is important to capture the sarcasm that is embedded in the OCR.

The paper made a further unique contribution by conceptualizing and defining the perception score. This score is designed to overcome the customer biases in giving out star ratings and the discrepancies between the OCR content and the star rating. Extant research has reported that the sentiments expressed in the OCR are important predictors of product sales [45] whereas emotions embedded in the OCR impact the helpfulness of the review [17,32] or sales [31,49]. The perception score is an integrative quantitative measure reflecting both sentiments and emotions expressed in the OCR data in addition to reactions from emojis and can be an influential factor in the customers' decision to download a mobile app. It is an intrinsic measure of the perceptions expressed in the

Table 4
Regression analysis between opinions in the OCR content and star rating for the mobile payment apps.

		Model 1	Model 2	Model 3
Standardized coefficients				
Control variables	App	0.233***	0.143***	0.141***
	Word count	−0.179***	−0.115***	−0.11***
	Emoji count	0.014**	0.018***	0.014***
Main effects	Positive reaction in emojis		0.055***	0.057***
	Negative reaction in emojis		−0.069***	−0.069***
	Anger		−0.059***	−0.066***
	Anticipation		−0.101***	−0.079***
	Disgust		−0.054***	−0.067***
	Fear		0.041***	0.038***
	Happiness		0.202***	0.182***
	Sadness		−0.039***	−0.052***
	Surprise		0.013*	0.025***
	Trust		−0.12***	−0.112***
	Consolidated sentiment		0.376***	0.486***
Interaction effects	Positive reaction in emojis X Consolidated sentiment			0.005
	Anticipation X Consolidated sentiment			−0.105***
	Happiness X Consolidated sentiment			0.078***
	Surprise X Consolidated sentiment			−0.033***
	Trust X Consolidated sentiment			−0.038***
	Negative reaction in emojis X Consolidated sentiment			−0.015*
	Anger X Consolidated sentiment			0.034***
	Disgust X Consolidated sentiment			−0.032***
	Fear X Consolidated sentiment			−0.05***
	Sadness X Consolidated sentiment			−0.049***
Model summary	Significance	<0.001	<0.001	<0.001
	Adjusted R^2	0.113	0.322	0.335
	ΔR^2		0.210***	0.013***

Note: Dependent variable is star rating. $N = 48{,}624$.
***: *p*-value <0.001, **: p-value <0.01, *: p-value <0.05.

review content and eliminates the element of subjectivity that is inherent in the star rating scale. Therefore, it is a richer representation of the customers' opinion about the mobile app.

The test of robustness showed that the results obtained in this research were consistent during the COVID-19 pandemic. This is important as the usage of mobile payment apps has significantly increased after the onset of the pandemic. At the same time, several negative emotions have become much stronger during the pandemic. Our results showed that the established relationships (e.g., between the opinions in the OCR and the star ratings) have become neither stronger or weaker during the pandemic. This result would be of interest to researchers who are studying the impact of the pandemic on human emotions and their subsequent impact on the use of mobile apps.

5.2. Practical contributions

OCR and star ratings influence purchase decisions in e-commerce [50]. Marketers have endeavored to maintain high star ratings for apps to boost their adoption [51]. However, they have little control over the star rating provided by the customers that do not closely reflect the textual content of the OCR. This has made the management of star ratings a challenge for the marketers and has made the star ratings unreliable for consumer decision making. The perception score that we proposed in this paper considered the emotions, sentiments, and reactions expressed in the textual content of the OCR. It offered a faster mechanism of identifying incongruent review ratings, as evidenced in

Table 5
A comparison of deviation from the constituents of opinions for star ratings and for perception scores.

	MSD_{Rating}	$MSD_{Perception\ score}$	MAD_{Rating}	$MAD_{Perception\ score}$
Consolidated sentiment	9.060	7.971	2.793	2.782
Anger	8.879	7.790	2.623	2.599
Anticipation	6.599	5.510	2.180	2.092
Disgust	9.696	8.607	2.784	2.778
Fear	9.077	7.988	2.658	2.635
Happiness	6.812	5.723	2.254	2.295
Sadness	9.074	7.984	2.642	2.615
Surprise	7.937	6.848	2.472	2.451
Trust	6.252	5.163	2.123	2.006
Positive reaction in emojis	349,422.167	349,421.078	45.504	45.501
Negative reaction in emojis	144,315.199	144,314.110	30.457	30.454

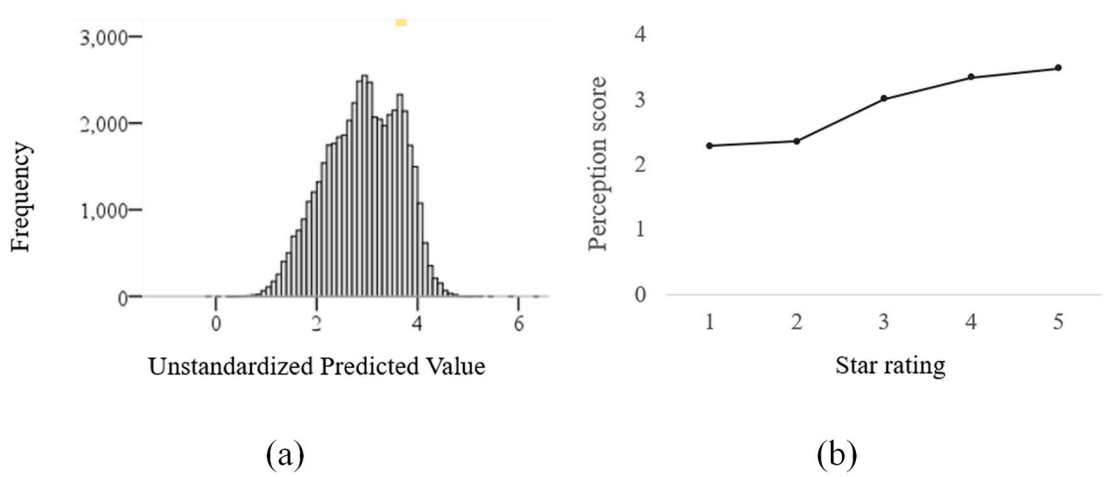

Fig. 6. (a) Histogram of the perception score. (b) The relationship between perception score and star ratings.

Table 6
A comparison of star rating and perception score for a sample of the OCR.

Sample OCR	Star rating	Perception score
Excellent app. I'm very happy with this app 😀😊👍😄🙂👏😀😊👍😄🙂👏😀😊👍😄🙂👏😀😊	5	5
Was using this app for so long but recently having problem with scanning QR code. It didn't take any entry. Am just so frustrated because I have to add other no. Just to pay. Such a hassle.	5	1.851
Very nicely done ✅ *quickly* 😊 *payment best quality* 👍😊👍😊	1	4.735
Only few starting transactions received a cashback then rewarded by some unrelated coupons Hopeless app	1	2.996
Very bad experience, payment stuck once so I have to make another it also stuck one failed after some hours but another one it still processing. Please cancel the update 😭😭😭😭😭😭😭😭😭 *I will never use this app again.*	1	0

Table 7
Regression analysis between opinions in the OCR content and star rating for the mobile payment apps during the COVID-19 pandemic.

		Model 4	Model 5
Standardized coefficients			
	App	0.121***	
Control variables	Word count	−0.114***	−0.163***
	Emoji count	0.013**	0.011*
	Positive reaction in emojis	0.054***	0.052***
	Negative reaction in emojis	−0.067***	−0.063***
	Anger	−0.066***	−0.074***
	Anticipation	−0.084***	−0.066***
	Disgust	−0.068***	−0.066***
	Fear	0.037***	0.048***
	Happiness	0.182***	0.191***
	Sadness	−0.058***	−0.048***
	Surprise	0.028***	0.029***
	Trust	−0.12***	−0.114***
Main effects	Consolidated sentiment	0.528***	0.493***
	Positive reaction in emojis X Consolidated sentiment	0.005	0.002
	Anticipation X Consolidated sentiment	−0.118***	−0.07***
	Happiness X Consolidated sentiment	0.096***	0.025*
	Surprise X Consolidated sentiment	−0.021**	−0.04***
	Trust X Consolidated sentiment	−0.063***	−0.016
	Negative reaction in emojis X Consolidated sentiment	−0.015*	−0.013
	Anger X Consolidated sentiment	0.028***	0.039***
	Disgust X Consolidated sentiment	−0.039***	−0.018**
Interaction effects	Fear X Consolidated sentiment	−0.047***	−0.062***
	Sadness X Consolidated sentiment	−0.063***	−0.028***
	Significance	<0.001	<0.001
Model summary	Adjusted R^2	0.363	0.306
	Number of reviews (N)	40,679	39,786

Note: Dependent variable is star rating.
***: *p*-value <0.001, **: *p*-value <0.01, *: *p*-value <0.05.

case of fake reviews. It provided the marketer an alternate metric to display to the interested users. Also, it could be used to compare the positivity of two reviews and sort reviews according to their positivity, as explained earlier in Fig. 7. The perception score can be expected to reduce the marketers' dependency on the subjective star ratings.

Marketers may apply the text mining techniques used in this research to understand the mindset of the customers when using the apps. This is expressed through the relative impact of the three dimensions of opinion ($\text{Sentiment}_{\text{Text}}$, $\text{Emotion}_{\text{Text}}$, and $\text{Reaction}_{\text{Emoji}}$) in the OCR on app ratings. Our research shows that marketers and data analysts should learn about the rich information content in emojis and include them in the analysis of customer reviews as reactions from emojis can provide an additional dimension for capturing the opinion of the customers.

The perception score introduces new possibilities to organizing the OCR data on the sites. With the present system, customers have to scroll through the page to find the information they are looking for. Organizing the reviews by perception is expected to remove the barrier introduced by colocation of inconsistent reviews and improve navigability to desired information. Viewers will gain the ability to locate the positive, negative and tepid reviews as per their choice. It will reduce customers' time and effort in scavenging for information, and ease the users' app download decision, and consequently increase app usage. The visual tool to summarize the perception in the review of a mobile payment app can empower the less IT savvy customers in India with low level of literacy to make judicious decision about the selection of a mobile app for financial payments. In fact, a past study has shown that color-coded graphical user interface works well in a low-literacy multi-language environment [52]. A visual tool to present the perception in a review will benefit the non-English speaking audience as well. The empowerment is expected to motivate app usage and smartphone adoption. It will improve information exchange, for the masses at the bottom of the pyramid, particularly for rural women who fall behind in term of education, and employment [53]. An increase in the adoption of mobile payment apps can also act as a catalyst for greater financial inclusion and pave the way for a digital less-cash economy in India.

5.3. Limitations and future research directions

We studied reviews of two leading mobile payment Android apps in India. We limited the study to the Android platform to maintain the homogeneity of the reviewers. Further research is needed for apps on the iOS, other payment apps in different geographical locations, and apps providing other digital services. We used Plutchik's [6] classification of basic emotions. In future, researchers may use other emotion classification theories from psychology, such as the pleasure-arousal-dominance model for categorizing emotions in the OCR. We used existing data mining tools and techniques for the analysis. However, the field of text mining is maturing, and researchers may use newer techniques such as deep learning to analyze the OCR. Researchers may also consider how to segregate fake reviews and conduct the analysis on genuine OCR only [54]. Companies like Amazon are working on identification of fake reviews [55]. Researchers may avail reviews data from such organizations to prevent the analysis of fake reviews.

In this paper, we proposed the perception score, and verified the impact of the score on constituent opinions statistically. Further research is needed to test the usability and customer acceptability of the new score. Customers are accustomed to star ratings. Any changes of a system are expected to impact its perceived usefulness and perceived ease of use [12]. Further research is needed to empirically determine if

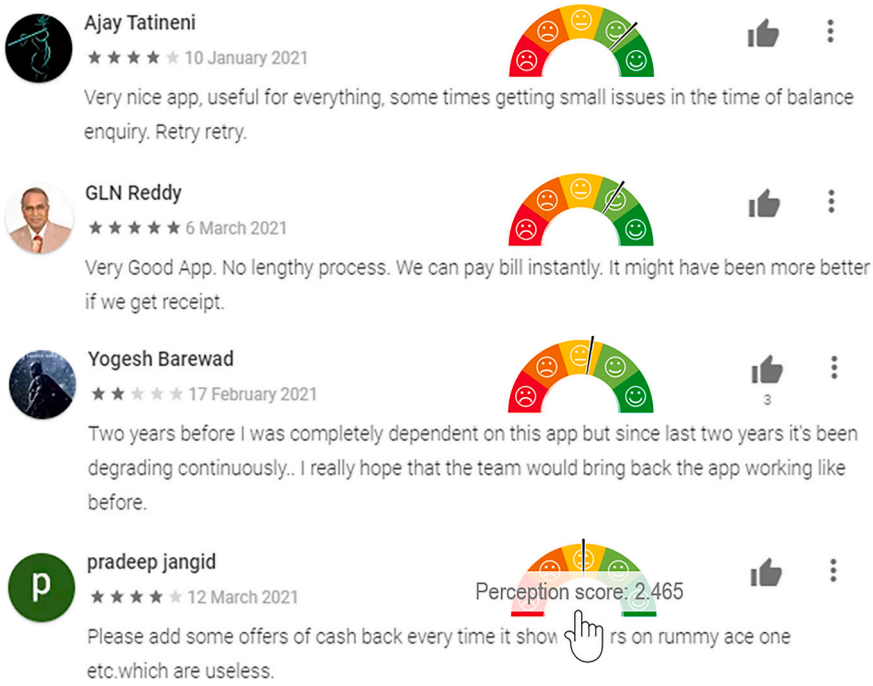

Fig. 7. A proposed arrangement of the OCR using the perception score.

the customers find the perception score more useful than star ratings through customer surveys and laboratory experiments. We have provided a possible presentation scheme for the perception score in Fig. 7. App developers and designers may refine the presentation. Further research is needed to study the influence of such presentation schemes on app downloads and customer satisfaction.

6. Conclusion

In this paper, we explored the sentiments, emotions, and reactions in the OCR of mobile apps and their impact on star ratings. The analysis showed a significant influence of customers' opinions mined from the rich content of the OCR on star ratings for the apps. The analysis highlighted the importance of including emojis while extracting opinions from the OCR. We devised the perception score that could reflect the constituent sentiments, emotions, and reactions better than the star ratings. We showed how sorting the reviews according to a visual depiction of the perception score can benefit the less educated smartphone users. App marketers may use the perception score to effectively assess the opinions expressed in individual reviews and summarize market reactions to the apps to enhance their popularity.

References

[1] M.A. Harris, R. Brookshire, A.G. Chin, Identifying factors influencing consumers' intent to install mobile applications, Int. J. Inf. Manag. 36 (2016) 441–450, https://doi.org/10.1016/j.ijinfomgt.2016.02.004.
[2] C.Z. Liu, Y.A. Au, H.S. Choi, Effects of freemium strategy in the mobile app market: an empirical study of Google play, J. Manag. Inf. Syst. 31 (2014) 326–354, https://doi.org/10.1080/07421222.2014.995564.
[3] G. Iyer, D. Kuksov, Consumer feelings and equilibrium product quality, J. Econ. Manag. Strateg. 19 (2010) 137–168, https://doi.org/10.1111/j.1530-9134.2009.00248.x.
[4] N. Vila, I. Kuster, Consumer feelings and behaviours towards well designed websites, Inf. Manag. 48 (2011) 166–177, https://doi.org/10.1016/j.im.2011.04.003.
[5] S. Chatterjee, Explaining customer ratings and recommendations by combining qualitative and quantitative user generated contents, Decis. Support. Syst. 119 (2019) 14–22, https://doi.org/10.1016/j.dss.2019.02.008.
[6] R. Plutchik, A General Psychoevolutionary Theory of Emotion, Academic Press Inc., New York, 1980, https://doi.org/10.1016/b978-0-12-558701-3.50007-7.
[7] R. Ullah, N. Amblee, W. Kim, H. Lee, From valence to emotions: exploring the distribution of emotions in online product reviews, Decis. Support. Syst. 81 (2016) 41–53, https://doi.org/10.1016/j.dss.2015.10.007.
[8] D. Yin, S.D. Bond, H. Zhang, Anxious or angry? Effects of discrete emotions on the perceived helpfulness of online reviews, MIS Q. 38 (2014) 539–560. https://www.jstor.org/stable/26634939.
[9] How these Cornell Tech Students Created a Mobile Operating System to Help Low-Literacy Users, Cornell Tech, 2018. https://tech.cornell.edu/news/how-these-cornell-tech-students-created-a-mobile-operating-system-to-help-low-literacy-users/.
[10] Newzoo, Top Countries by Smartphone Users. https://newzoo.com/insights/rankings/top-countries-by-smartphone-penetration-and-users/, 2021 (accessed June 24, 2021).
[11] Paypers, Exclusive Interview with NPCI – Instant Payments and Digital Transformation in India. https://thepaypers.com/interviews/exclusive-interview-with-npci-instant-payments-and-digital-transformation-in-india–1246856, 2021 (accessed June 24, 2021).
[12] S.K. Sharma, Integrating cognitive antecedents into TAM to explain mobile banking behavioral intention: a SEM-neural network modeling, Inf. Syst. Front. 21 (2019) 815–827, https://doi.org/10.1007/s10796-017-9775-x.
[13] H.C. Yang, Bon appétit for apps: young American consumers' acceptance of mobile applications, J. Comput. Inf. Syst. 53 (2013) 85–96, https://doi.org/10.1080/08874417.2013.11645635.
[14] V.M. Wottrich, E.A. van Reijmersdal, E.G. Smit, The privacy trade-off for mobile app downloads: the roles of app value, intrusiveness, and privacy concerns, Decis. Support. Syst. 106 (2018) 44–52, https://doi.org/10.1016/j.dss.2017.12.003.
[15] F. Liébana-Cabanillas, J. Sánchez-Fernández, F. Muñoz-Leiva, The moderating effect of experience in the adoption of mobile payment tools in virtual social networks: the m-payment acceptance model in virtual social networks (MPAM-VSN), Int. J. Inf. Manag. 34 (2014) 151–166, https://doi.org/10.1016/j.ijinfomgt.2013.12.006.
[16] C. Xu, D. Peak, V. Prybutok, A customer value, satisfaction, and loyalty perspective of mobile application recommendations, Decis. Support. Syst. 79 (2015) 171–183, https://doi.org/10.1016/j.dss.2015.08.008.
[17] A. Felbermayr, A. Nanopoulos, The role of emotions for the perceived usefulness in online customer reviews, J. Interact. Mark. 36 (2016) 60–76, https://doi.org/10.1016/j.intmar.2016.05.004.
[18] E.W. Anderson, Customer satisfaction and word of mouth, J. Serv. Res. 1 (1998) 5–17, https://doi.org/10.1177/109467059800100102.
[19] J.A. Chevalier, D. Mayzlin, The effect of word of mouth on sales: online book reviews, J. Mark. Res. 43 (2006) 345–354.
[20] K. Xu, S.S. Liao, J. Li, Y. Song, Mining comparative opinions from customer reviews for competitive intelligence, Decis. Support. Syst. 50 (2011) 743–754, https://doi.org/10.1016/j.dss.2010.08.021.
[21] Y.Y.Y. Chan, E.W.T. Ngai, Conceptualising electronic word of mouth activity: an input-process-output perspective, Mark. Intell. Plan. 29 (2011) 488–516, https://doi.org/10.1108/02634501111153692.
[22] X. Xu, X. Wang, Y. Li, M. Haghighi, Business intelligence in online customer textual reviews: understanding consumer perceptions and influential factors, Int. J. Inf. Manag. 37 (2017) 673–683, https://doi.org/10.1016/j.ijinfomgt.2017.06.004.

[23] J. Zhang, S. Piramuthu, Product recommendation with latent review topics, Inf. Syst. Front. 20 (2018) 617–625, https://doi.org/10.1007/s10796-016-9697-z.
[24] J. Guo, X. Wang, Y. Wu, Positive emotion bias: role of emotional content from online customer reviews in purchase decisions, J. Retail. Consum. Serv. 52 (2020), 101891, https://doi.org/10.1016/j.jretconser.2019.101891.
[25] S. Karimi, F. Wang, Online review helpfulness: impact of reviewer profile image, Decis. Support. Syst. 96 (2017) 39–48, https://doi.org/10.1016/j.dss.2017.02.001.
[26] A.A. Alalwan, Mobile food ordering apps: an empirical study of the factors affecting customer e-satisfaction and continued intention to reuse, Int. J. Inf. Manag. 50 (2020) 28–44, https://doi.org/10.1016/j.ijinfomgt.2019.04.008.
[27] W. Tafesse, The effect of app store strategy on app rating: the moderating role of hedonic and utilitarian mobile apps, Int. J. Inf. Manag. 57 (2021), 102299, https://doi.org/10.1016/j.ijinfomgt.2020.102299.
[28] P. Zhang, Y. Ge, H.M. Lee, E-WOM's impact on app development, J. Comput. Inf. Syst. 60 (2020) 418–427, https://doi.org/10.1080/08874417.2018.1517283.
[29] S. Zhou, Z. Qiao, Q. Du, G.A. Wang, W. Fan, X. Yan, Measuring customer agility from online reviews using big data text analytics, J. Manag. Inf. Syst. 35 (2018) 510–539, https://doi.org/10.1080/07421222.2018.1451956.
[30] E. Ismagilova, E.L. Slade, N.P. Rana, Y.K. Dwivedi, The effect of electronic word of mouth communications on intention to buy: a meta-analysis, Inf. Syst. Front. 22 (2020) 1203–1226, https://doi.org/10.1007/s10796-019-09924-y.
[31] N. Hu, N.S. Koh, S.K. Reddy, Ratings lead you to the product, reviews help you clinch it? The mediating role of online review sentiments on product sales, Decis. Support. Syst. 57 (2014) 42–53, https://doi.org/10.1016/j.dss.2013.07.009.
[32] G. Ren, T. Hong, Examining the relationship between specific negative emotions and the perceived helpfulness of online reviews, Inf. Process. Manag. 56 (2019) 1425–1438, https://doi.org/10.1016/j.ipm.2018.04.003.
[33] R.P. Bagozzi, M. Gopinath, P.U. Nyer, The role of emotions in marketing, J. Acad. Mark. Sci. 27 (1999) 184–206, https://doi.org/10.1177/0092070399272005.
[34] C. Ranaweera, K. Menon, For better or for worse?: adverse effects of relationship age and continuance commitment on positive and negative word of mouth, Eur. J. Mark. 47 (2013) 1598–1621, https://doi.org/10.1108/EJM-06-2011-0295.
[35] S. Djamasbi, D.M. Strong, M. Dishaw, Affect and acceptance: examining the effects of positive mood on the technology acceptance model, Decis. Support. Syst. 48 (2010) 383–394, https://doi.org/10.1016/j.dss.2009.10.002.
[36] R.C. Ho, M.S. Withanage, K.W. Khong, Sentiment drivers of hotel customers: a hybrid approach using unstructured data from online reviews, Asia-Pacific J. Bus. Adm. 12 (2020) 237–250, https://doi.org/10.1108/APJBA-09-2019-0192.
[37] Y. Choi, H. Lee, Data properties and the performance of sentiment classification for electronic commerce applications, Inf. Syst. Front. 19 (2017) 993–1012, https://doi.org/10.1007/s10796-017-9741-7.
[38] N.Y. Lee, S.M. Noble, A.R. Zablah, So distant, yet useful: the impact of distal stories on customers' service expectations, J. Bus. Res. 113 (2020) 230–242, https://doi.org/10.1016/j.jbusres.2020.01.044.
[39] H.P. Shih, P.C. Sung, Addressing the review-based learning and private information approaches to foster platform continuance, Inf. Syst. Front. (2020), https://doi.org/10.1007/s10796-020-09985-4. Advance online publication.
[40] A. Subroto, M. Christianis, Rating prediction of peer-to-peer accommodation through attributes and topics from customer review, J. Big Data. 8 (2021), https://doi.org/10.1186/s40537-020-00395-6.
[41] Y. Zhao, X. Xu, M. Wang, Predicting overall customer satisfaction: big data evidence from hotel online textual reviews, Int. J. Hosp. Manag. 76 (2019) 111–121, https://doi.org/10.1016/j.ijhm.2018.03.017.
[42] J. Ge, U. Gretzel, Emoji rhetoric: a social media influencer perspective, J. Mark. Manag. 34 (2018) 1272–1295, https://doi.org/10.1080/0267257X.2018.1483960.
[43] NPCI, NPCI UPI Statistics, NPCI UPI Stat. https://www.npci.org.in/what-we-do/upi/product-statistics, 2022 (accessed February 25, 2021).
[44] A.K. Rathore, P.V. Ilavarasan, Pre- and post-launch emotions in new product development: insights from twitter analytics of three products, Int. J. Inf. Manag. 50 (2020) 111–127, https://doi.org/10.1016/j.ijinfomgt.2019.05.015.
[45] A.Y.L. Chong, B. Li, E.W.T. Ngai, E. Ch`ng, F. Lee, Predicting online product sales via online reviews, sentiments, and promotion strategies: a big data architecture and neural network approach, Int. J. Oper. Prod. Manag. 36 (2016) 358–383, https://doi.org/10.1108/IJOPM-03-2015-0151.
[46] J.R. Landis, G.G. Koch, The measurement of observer agreement for categorical data, Biometrics. 33 (1977) 159–174, https://doi.org/10.2307/2529310.
[47] N. Hu, J. Zhang, P.A. Pavlou, Overcoming the J-shaped distribution of product reviews, Commun. ACM 52 (2009) 144–147, https://doi.org/10.1145/1562764.1562800.
[48] S. Sangwan, M.A. Akhtar, P. Behera, A. Ekbal, Exploring multimodality for sarcasm detection, in: Proceedings of the Int. Jt. Conf. Neural Networks, Glasgow, UK, 2020, p. 7.
[49] H. Hong, D. Xu, G.A. Wang, W. Fan, Understanding the determinants of online review helpfulness: a meta-analytic investigation, Decis. Support. Syst. 102 (2017) 1–11, https://doi.org/10.1016/j.dss.2017.06.007.
[50] J. Gao, C. Zhang, K. Wang, S. Ba, Understanding online purchase decision making: the effects of unconscious thought, information quality, and information quantity, Decis. Support. Syst. 53 (2012) 772–781, https://doi.org/10.1016/j.dss.2012.05.011.
[51] A.P. Kapoor, M. Vij, How to boost your app store rating? An empirical assessment of ratings for mobile banking apps, J. Theor. Appl. Electron. Commer. Res. 15 (2020) 99–115, https://doi.org/10.4067/S0718-18762020000100108.
[52] H. Shah, A. Sengupta, Designing mobile based computational support for low-literate community health workers, Int. J. Hum. Comput. Stud. 115 (2018) 1–8, https://doi.org/10.1016/j.ijhcs.2018.01.007.
[53] W. Ma, R.Q. Grafton, A. Renwick, Smartphone use and income growth in rural China: empirical results and policy implications, Electron. Commer. Res. 20 (2020) 713–736, https://doi.org/10.1007/s10660-018-9323-x.
[54] Y. Wu, E.W.T. Ngai, P. Wu, C. Wu, Fake online reviews: literature review, synthesis, and directions for future research, Decis. Support. Syst. 132 (2020), 113280, https://doi.org/10.1016/j.dss.2020.113280.
[55] D. Proserpio, B. Hollenbeck, S. He, How fake customer reviews do - and don't - work, Harv. Bus. Rev. (2020) https://hbr.org/2020/11/how-fake-customer-reviews-do-and-dont-work.

Suparna Dhar is an Associate Professor of Computing & Analytics at the NSHM Knowledge Campus Kolkata. She has completed her Bachelor of Science in Mathematics and Masters of Statistics from Indian Statistical Institute, Kolkata. Her research interests are in machine learning and AI, social networking, blockchain, fintech, and Cybersecurity. Her research has been published in journals like *Journal of Organizational Computing and Electronic Commerce* as well as those of conferences like ICIS and *Workshop on e-Business* (pre-ICIS).

Indranil Bose is Distinguished Professor of Management Information Systems at the NEOMA Business School. He acts as Head of the Area of Excellence in Artificial Intelligence, Data Science, and Business. He holds a BTech from the Indian Institute of Technology, MS from the University of Iowa, and MS and PhD from Purdue University. His research interests are in business analytics, digital transformation, information security, and management of emerging technologies. His publications have appeared in MIS Quarterly, Journal of the MIS, Communications of the ACM, Communications of the AIS, Computers and Operations Research, Decision Support Systems, Electronic Markets, European Journal of Operational Research, Information & Management, International Journal of Production Economics, Journal of Organizational Computing and Electronic Commerce, Journal of the American Society for Information Science and Technology, Operations Research Letters, Technological Forecasting and Social Change, etc. He serves as Senior Editor of Decision Support Systems and Pacific Asia Journal of the AIS, and as Associate Editor of Communications of the AIS, Information & Management, and Journal of the AIS.